

Microteaching Quantum Computing

Course Instructor:

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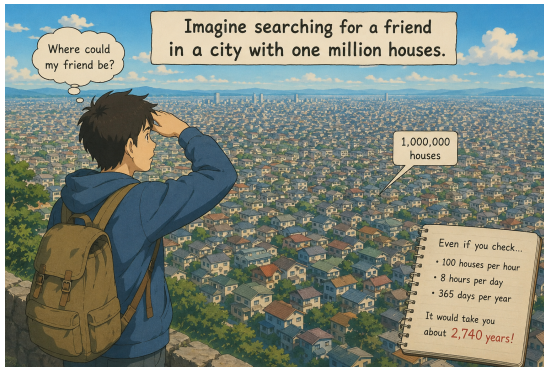
Vignan's Foundation for Science, Technology and Research

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Learning Outcomes

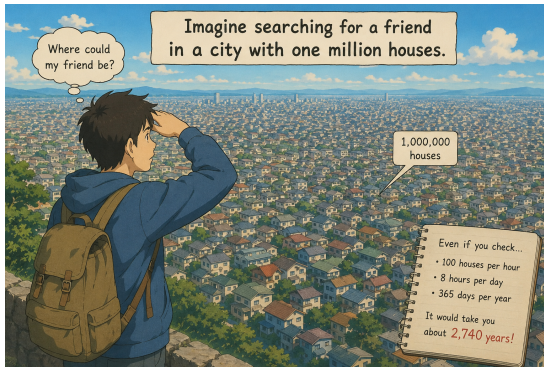
- Explain a classical bit.
- Describe a qubit.
- Understand superposition qualitatively.
- Mention one application of quantum computing.

Motivation – A thought experiment



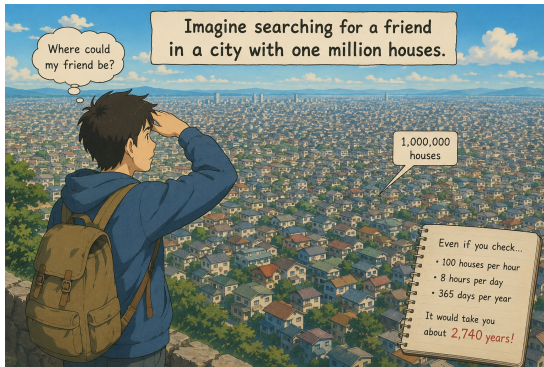
Imagine searching for a friend in a city with one million houses.

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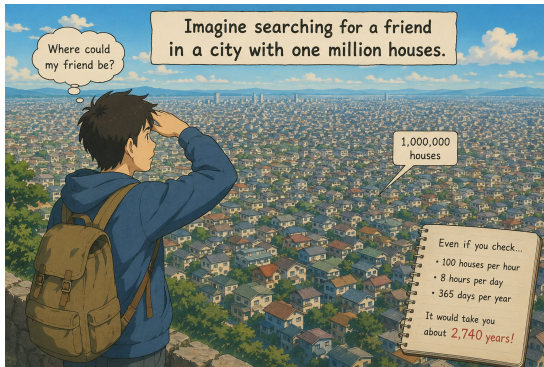
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Can Nature help us compute differently?

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Imagine searching for a friend in a city with one million houses. A classical computer checks one possibility after another. Can Nature help us compute differently?

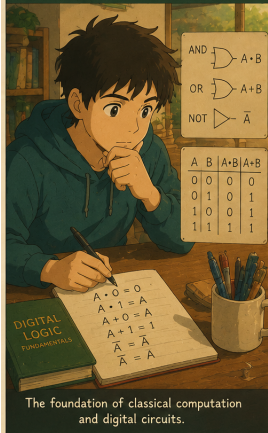
Today's Question

What makes a quantum computer different?

Prerequisites

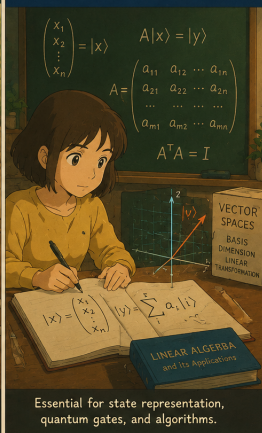
1. BOOLEAN ALGEBRA

The language of logic



2. LINEAR ALGEBRA

The mathematics of vectors and transformations



3. QUANTUM MECHANICS

The physical laws of the quantum world



These three pillars support the beautiful edifice of Quantum Computing. 

Boolean Algebra: Classical Bit



A classical bit has only two states:

0 or 1

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Examples:

- Light switch: ON/OFF
- Memory cell
- Voltage High/Low

Boolean Algebra: Classical Bit



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Question: Can a classical bit be both 0 and 1 simultaneously?

Linear Algebra: Quantum Bit (Qubit)

A qubit is the basic unit of quantum information.

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Unlike a classical bit, a qubit can exist in a **superposition** of states.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

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$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Measurement gives either 0 or 1.

Superposition Analogy

Classical Coin

- Heads
- Tails

Spinning Coin

- Appears to show both possibilities
- Only an analogy

Superposition Analogy

Classical Coin

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A qubit is fundamentally a quantum object—not literally a spinning coin.

Spinning Coin

- Appears to show both possibilities
- Only an analogy

Why Quantum Computing?

Eratosthenes' Sieve Method
To find primes and factorize numbers

1. Write numbers from 2 to n .

2	3	4	5	6	7	8	9	10	
11	12	13	14	15	16	17	18	19	20

2. Circle the first number (2).
Cross out all multiples of 2.

2	3	4	5	6	7	8	9	10	
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3. Circle the next uncrossed number (3).
Cross out all multiples of 3.

2	3	4	5	6	7	8	9	10	
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4. Circle the next uncrossed number (5).
Cross out all multiples of 5.

2	3	4	5	6	7	8	9	10	
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5. Continue until n .
The uncrossed numbers are primes.

2	3	4	5	6	7	8	9	10	
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Primes up to 20:
2, 3, 5, 7, 11, 13, 17, 19

Using Primes for Factorization
Break a number into its prime factors using the primes from the sieve.

Example 1: Factorize 84

2	84
2	42
3	21
7	7
	1

$84 = 2 \times 2 \times 3 \times 7$
 $= 2^2 \times 3 \times 7$

Use the smallest prime each time!

Example 2: Factorize 360

2	360
2	180
2	90
3	45
3	15
5	3
	1

$360 = 2 \times 2 \times 2 \times 3 \times 3 \times 5$
 $= 2^3 \times 3^2 \times 5$

Why it works?
By crossing out multiples, we are left with numbers that have no smaller prime factors – these are primes!

Potential applications:

- Cryptography
- Drug discovery
- Material science
- Optimization
- Quantum simulation

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Important: Quantum computers are *not* faster for every problem.

Think–Pair–Share

Two classical bits produce:

00, 01, 10, 11

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Two classical bits produce:

00, 01, 10, 11

Question:

What changes if we replace them with two qubits?

Discuss with your neighbour for one minute.

Summary

- Classical bit: 0 or 1.
- Qubit: quantum state.
- Superposition is a uniquely quantum property.
- Quantum computing enables new algorithms for selected problems.

Exit Question:

Name one application of quantum computing.

Thank You

Questions?